

Sorwatom — Feature Documentation

From: June 19, 2026 (day one) → **To:** June 30, 2026 (today)

Site: sorwatom.rf.gd · InfinityFree hosting · PHP + MySQL

1. Site Architecture

File Structure

```

public_html/
├── index.php           Homepage
├── about.php          Our Story page
├── products.php       Products listing
├── product-ketchup.php Ketchup detail
├── product-tomato-paste.php Tomato Paste detail
├── product-masala.php  Pilau Masala detail
├── product-vinegar.php Pure Vinegar detail
├── blog.php           Journal listing
├── blog-post.php      Single post view
├── contact.php        Contact form
├── privacy.php        Privacy policy
├── feed.php           RSS feed (/feed)
├── 404.php            Custom error page
├── favicon.ico        Browser tab icon (root)
├──
├── partials/          Shared PHP partials
│   ├── _head.php      <head> meta, CSS links, lang bootstrap
│   ├── _scripts.php   JS includes
│   ├── nav.php        Site navigation
│   ├── footer.php     Footer
│   └── _lang.php       i18n session + __t() / __r() helpers
├──
├── lang/              Translation files
│   ├── en.php         English (282 lines)
│   ├── fr.php         French (265 lines)
│   └── sw.php          Kiswahili (262 lines)
├──
├── data/
│   └── blog.php        All blog DB query functions
├──
├── config/
│   ├── app.php        Constants (DB, SITE_URL, admin hash)
│   └── database.php    PDO singleton (get_db())
├──
├── admin/             Admin panel (auth-gated)
│   ├── index.php      Dashboard
│   ├── login.php / logout.php
│   ├── posts.php      Post list
│   └── post-edit.php   Create / edit post (WYSIWYG)

```

```

├── post-delete.php      Delete handler
├── setup.php            One-time DB table creator
├── image-upload.php     AJAX image upload endpoint
├── db-check.php         Diagnostic tool
├── _auth.php            Auth guard
├── _nav.php             Admin nav partial
├── assets/admin.css     Admin stylesheet (610 lines)
├── assets/
│   ├── css/
│   │   ├── tokens.css  Design tokens
│   │   ├── reset.css   CSS reset
│   │   ├── layout.css  Grid / containers
│   │   ├── global.css  Base typography
│   │   ├── components.css Shared components (1 362 lines)
│   │   └── pages/      Per-page stylesheets
│   │       ├── home.css
│   │       ├── about.css
│   │       ├── products.css
│   │       ├── blog.css
│   │       └── contact.css
│   └── img/
│       ├── products/   PNG + WebP product images
│       ├── slider/     PNG/JPG + WebP hero images
│       └── blog/        Uploaded blog post images

```

Clean URL Routing (`.htaccess`)

Browser URL	Serves
/	index.php
/about	about.php
/products	products.php
/product-ketchup	product-ketchup.php
/blog	blog.php
/blog/my-post-slug	blog-post.php?slug=my-post-slug
/contact	contact.php
/feed	feed.php
/admin/posts	admin/posts.php

POST requests are excluded from the clean-URL redirect (a 301 would convert POST → GET and lose form data).

2. Design System & CSS

Tokens (`tokens.css`)

- **Colour palette:** `--col-ground` (dark green), `--col-accent` (amber), `--col-surface-*`, `--col-text-*`, `--col-border`
- **Typography scale:** fluid `--step--1` → `--step-5` using `clamp()`
- **Spacing scale:** `--space-xs` → `--space-3xl`
- **Motion:** `--dur-fast`, `--dur-base`, `--ease-out`
- **Radius:** `--radius-sm`, `--radius-lg`

Component Library (`components.css` — 1 362 lines)

- **Hero:** `position: relative; display: flex; align-items: flex-end`. Variants: `hero--full` (100svh), `hero--half`. Background via `hero__bg` (absolute `<img>`) or `hero__bg--pattern` (CSS radial gradient). Overlay via `::after` pseudo-element
- **Navigation:** fixed top bar, logo, links, mobile hamburger overlay with accessibility (focus management on close)
- **Buttons:** `.btn-primary`, `.btn-ghost`, `.btn--text`, `.btn-delete`
- **Cards:** post cards, product cards, related cards
- **Badges:** `.badge`, `.badge--outline`
- **Stats bar:** founded year, countries, products count
- **Author bio:** avatar (image or initial fallback), name, bio text
- **Share buttons:** WhatsApp link + copy-link button
- **Pagination:** prev/next with page info
- **CTA block:** full-width dark promotional strip
- **Forms:** inputs, textareas, selects, labels, hints, error/success alerts

Performance

- All hero and product images available as **WebP** (`<picture>` element with PNG/JPEG fallback)
- `.htaccess` auto-serves `.webp` to supporting browsers without changing URLs
- `<link rel="preload">` hints on above-the-fold hero images
- **Image sizes reduced ~10x:** hero images from 2.4–3.1 MB → 156–232 KB per image
- Gzip/Deflate compression for HTML, CSS, JS, JSON, fonts
- Browser cache headers: CSS/JS 1 month · images 6 months · fonts 1 year

3. Public Pages

Homepage (`index.php`)

- Full-viewport hero with WebP background image, headline, subtitle, CTA buttons
- About teaser section
- Product grid (4 products)
- Wholesale / contact CTA block
- All strings translatable via `__t()` / `__r()`

About (`about.php`)

- Full-viewport hero with stats bar pinned to the bottom (Founded · Countries · Products) — uses `margin-top: auto` on a flex-column hero
- Brand story section
- Team cards (photo, name, role, bio)
- "Our Journey" narrative / product promise section
- Top clearance: `clamp(9rem, 20svh, 15rem)` to clear the fixed nav on all viewports

Products (`products.php`)

- Product grid with images, names, and links to detail pages

Product Detail Pages (×4)

Each of `product-ketchup.php`, `product-tomato-paste.php`, `product-masala.php`, `product-vinegar.php` has:

- Hero with eyebrow label and styled title
- Product image (transparent PNG, `object-fit: contain`)
- Category badge, full title, description
- Feature bullet list (4 items with checkmarks)
- CTA buttons: "Request a Quote →" and "View All Products"
- Related products grid (3 cards)
- All strings translated (EN / FR / SW) via `pd.*` keys

Blog Listing (`blog.php`)

- Hero with pattern background, eyebrow, title, subtitle
- Category filter pills (All + per-category)
- Featured post card (large, `16/9` aspect ratio)
- Post grid (up to 9 per page, `16/9` cards)
- Reading time shown on each card (calculated at 200 wpm)
- Empty-state message when no posts in category
- Pagination: prev / next / "Page X of Y"
- All UI strings translated via `blog.*` keys

Blog Post (`blog-post.php`)

- Hero: `hero--full` with `min-height: 56svh`, category badge, post title, author · date
- Featured image (displayed at top of article body)
- Tags row (badge pills)
- Post content — raw HTML output from DB (supports all rich text formatting)
- Share bar: WhatsApp share link + copy-link button (with "Copied!" feedback)
- Author bio card: avatar image or initial, name, bio
- "← Back to Journal" footer link
- Related posts grid (same category, up to 3)
- JSON-LD `Article` structured data (headline, description, datePublished, dateModified, author, publisher, image)
- Open Graph meta tags (title, description, image, URL, type=article)
- RSS feed link in `<head>`

- Copy-link JS uses PHP-rendered strings so "Copied!" / "Copy link" respect the active language

Contact ([contact.php](#))

- Contact form with AJAX submission and client-side validation

RSS Feed ([feed.php](#))

- Standard RSS 2.0 feed at [/feed](#)
- Lists all published posts with title, link, description, pubDate

404 Page ([404.php](#))

- Custom error page served via `.htaccess ErrorDocument 404 /404.php`

4. Multilingual System (i18n)

How it works

- Language stored in `$_SESSION['lang']`; default is `en`
- [lang-switch.php](#) handles the POST from the footer switcher and redirects back
- [partials/_lang.php](#) is bootstrapped by [_head.php](#) before any HTML — `__t()` and `__r()` are available on every page

Helper functions

Function	Returns	Use case
<code>__t(\$key)</code>	HTML-escaped string	Most strings
<code>__t(\$key, ['var' => 'val'])</code>	String with <code>{var}</code> replaced	Pagination info, dynamic counts
<code>__r(\$key)</code>	Raw HTML string	Titles containing <code></code> italic highlights

Supported languages

Code	Language
<code>en</code>	English
<code>fr</code>	Français
<code>sw</code>	Kiswahili

Translation coverage

Namespace	Pages covered
<code>nav.*</code>	All pages (navigation links)

Namespace	Pages covered
home.*	Homepage
about.*	About page
products.*	Products listing
pd.ketchup.*	Ketchup detail page
pd.tomato.*	Tomato Paste detail page
pd.masala.*	Masala detail page
pd.vinegar.*	Vinegar detail page
pd.rel.*	Related product labels (shared)
contact.*	Contact page
footer.*	Footer
privacy.*	Privacy policy
blog.*	Blog listing + post view

5. Blog System (Database-Driven)

Database Schema

blog_posts	id, slug, title, excerpt, content, category_id, author_id, image_url, status, published_at, created_at, updated_at
blog_categories	id, name, slug, color
blog_authors	id, name, bio, avatar_url
blog_post_tags	post_id, tag

Data Layer (data/blog.php — 218 lines)

Function	Purpose
blog_get_posts(\$page, \$per_page, \$category_slug)	Paginated post list for the listing page
blog_count_posts(\$category_slug)	Total count for pagination
blog_get_featured()	Latest published post for the featured card
blog_get_by_slug(\$slug)	Single post for the post view
blog_get_related(\$post_id, \$category_id)	Up to 3 related posts

Function	Purpose
<code>blog_get_categories()</code>	All categories with post counts
<code>blog_get_tags(\$post_id)</code>	Tags for a single post
<code>blog_save_post(\$data, \$id)</code>	Insert or update a post + replace tags
<code>blog_delete_post(\$id)</code>	Hard delete
<code>blog_get_all_posts_admin()</code>	Full list for admin table
<code>blog_get_post_admin(\$id)</code>	Single post for the edit form
<code>blog_get_authors()</code>	Author list for the select dropdown
<code>blog_get_all_categories_admin()</code>	Category list for the select dropdown
<code>blog_reading_time(\$html)</code>	Word count ÷ 200, minimum 1 min
<code>blog_hydrate(\$post)</code>	Attaches <code>reading_time</code> , <code>date_formatted</code> , <code>tags</code> to a post row
<code>blog_slugify(\$text)</code>	Converts text to a URL-safe slug

Database Connection (`config/database.php`)

- PDO singleton `get_db()` — one connection per request
- `charset=utf8mb4` in DSN
- `ATTR_ERRMODE => ERRMODE_EXCEPTION`
- `ATTR_EMULATE_PREPARES => false` (native prepared statements)

6. Admin Panel

Access & Security

- All admin pages begin with `require_once '_auth.php'` — unauthenticated requests redirect to `/admin/login`
- **Authentication:** bcrypt password via `password_verify()` against `$2y$` hash in `config/app.php`
- **CSRF protection:** token generated per session; every POST handler calls `csrf_verify()` — mismatch throws a 403
- `admin/.htaccess` — `Options -Indexes` (no directory listing)

Pages

Page	Function
<code>admin/index.php</code>	Dashboard — post count, category count, quick links
<code>admin/posts.php</code>	Post list table with edit/delete links
<code>admin/post-edit.php</code>	Create / edit post (WYSIWYG editor)
<code>admin/post-delete.php</code>	CSRF-verified delete handler

Page	Function
<code>admin/setup.php</code>	One-time DB table creation script
<code>admin/db-check.php</code>	Diagnostic: lists posts + image_url + disk check

Post Editor (`admin/post-edit.php` — 568 lines)

Fields saved to DB: title, slug, excerpt, content (HTML), category, author, image_url, status (draft/published), publish date, tags

Rich Text Editor (WYSIWYG)

- `contenteditable` div — no raw HTML tags visible to the user
- Toolbar buttons: **B** · *I* · U · ~~S~~ · H2 · H3 · P · Blockquote · Bullet List · Numbered List · Link · Insert Image · HR · Clear Formatting
- All formatting via `document.execCommand()`
- Toolbar highlights the active format as the cursor moves
- `onmousedown="return false"` on the toolbar prevents focus leaving the editor
- Tab key inserts 4 non-breaking spaces instead of tabbing away
- On form submit, content is synced to `<input type="hidden" name="content">` via a `submit` event listener

Featured Image Sidebar

- Drag-and-drop upload zone (HTML `<label>` over a hidden `<input type="file">`)
- Live preview with a × remove button after upload or URL entry
- URL text input as fallback (paste any image URL)
- After upload, the URL text input is updated automatically

Inline Image Insertion (modal)

- "Image" toolbar button opens a modal
- Cursor position is saved with `saveRange()` (`window.getSelection().getRangeAt(0).cloneRange()`) before the modal opens
- Supports image upload or URL entry, optional alt text
- On Insert, `restoreRange()` replaces the selection and `execCommand('insertHTML')` drops the `<img>` at the exact cursor position
- Escape key and backdrop click close the modal

Slug auto-generation

- Typing a title auto-generates a slug (`lowercase`, `hyphens`, `a-z0-9` only)
- Once the slug field is edited manually, auto-generation stops
- Slug preview shown inline: `/blog/your-post-slug`

Image Upload Endpoint (`admin/image-upload.php`)

- **Method:** POST (AJAX, `FormData`)
- **Auth:** requires active admin session + valid CSRF token in `FormData`

- **MIME validation:** `finfo_open(FILEINFO_MIME_TYPE)` — validates actual file bytes, not file extension
- **Allowed types:** JPEG, PNG, WebP, GIF
- **Size limit:** 8 MB
- **Filename:** `YYYYMMDD-<10 random hex chars>.ext`
- **Save path:** `assets/img/blog/`
- **Response:** `{"url": "/assets/img/blog/filename.ext"}` or `{"error": "..."}`

7. Footer

- 3-column grid: Brand (logo + tagline), Explore (nav links), HQ (address + contact)
- Social media links
- Copyright year from `date('Y')` — auto-increments annually
- "X+ years" heritage counter: `date('Y') - 1984` — also auto-increments
- Language switcher: **EN** · **FR** · **SW** buttons (POST to `lang-switch.php`)
- Logo shown in full colour (CSS `filter` removed)

8. Bug Fixes Applied (Day One → Today)

Bug	Root Cause	Fix
Mobile menu focus trap	Closing overlay left focus on <code>aria-hidden</code> element	Return focus to hamburger toggle before hiding overlay
Login form losing password	<code>.htaccess</code> 301 redirect converted POST → GET	Added <code>RewriteCond %{REQUEST_METHOD} !POST</code> to skip redirect on POST
Admin bcrypt mismatch	Python generates <code>\$2b\$</code> prefix; PHP expects <code>\$2y\$</code>	Converted hash prefix in <code>config/app.php</code>
Blog assets 404 at <code>/blog/slug</code>	Partials used relative paths; <code>/blog/slug</code> is one level deep	Changed all asset paths to absolute root paths (<code>/assets/...</code>)
Blog hero pattern dimmed	Overlay <code>::after</code> was applying opacity to the pattern div	Restored <code>opacity: 1</code> on <code>.hero__bg-pattern</code>
Blog card images not constrained	<code>max-height</code> on grid container doesn't shrink grid items	Added explicit height to <code>.img-wrap</code> + <code>object-fit: contain</code>
Logo pure white in footer/admin	<code>filter: brightness(0) invert(1)</code> in <code>components.css</code> (loaded globally)	Removed filter from <code>components.css</code> at the source
Fields not saving to DB	Delete <code><form></code> was nested inside the edit <code><form></code> ; its <code></form></code> closed the outer form early — <code>image_url</code> , <code>tags</code> , <code>category_id</code> , <code>author_id</code> were never submitted	Moved delete form outside the main form; delete button uses <code>form="delete-form"</code> attribute

Bug	Root Cause	Fix
Blog post hero content at top	<code>hero post-hero</code> class used — <code>post-hero</code> is undefined; <code>align-items: flex-end</code> not applied	Changed to <code>hero hero--full</code> ; added explicit <code>align-items</code> , increased <code>min-height</code> to <code>56svh</code> , added <code>padding-top: 5rem</code> to clear fixed nav
Favicon 404 on every page	Browsers request <code>/favicon.ico</code> at root regardless of <code><link rel="icon"></code>	Generated <code>favicon.ico</code> from <code>favicon.png</code> using Python (PNG-compressed ICO)